

Week 2 Guided Self-Study

- RISC-V State of the Union
 - Krste Asanović (RISC-V International)
- Arm vs x86 vs RISC-V: Does It Matter?
 - Jim Keller
- Watch: both videos (approximately 30 minutes total)
- Reflect: use the study questions on the next slide
- Focus: ISA design, software compatibility, and processor implementation

Study Questions

1. Why might two RISC-V processors not run the same binary? How do profiles help address this problem?
2. Why might a RISC ISA support variable-length instructions? Consider code density, future extensions, and decoding complexity.
3. Using this week's lecture, explain how two processors can implement the same ISA yet differ in forwarding, branch handling, and performance.
4. Why does backward compatibility make an ISA harder to change? Give one example from Keller's discussion.
5. Keller emphasizes processor implementation, while Asanović emphasizes the ISA ecosystem. How can both perspectives be valid? Support your answer with examples from the videos.